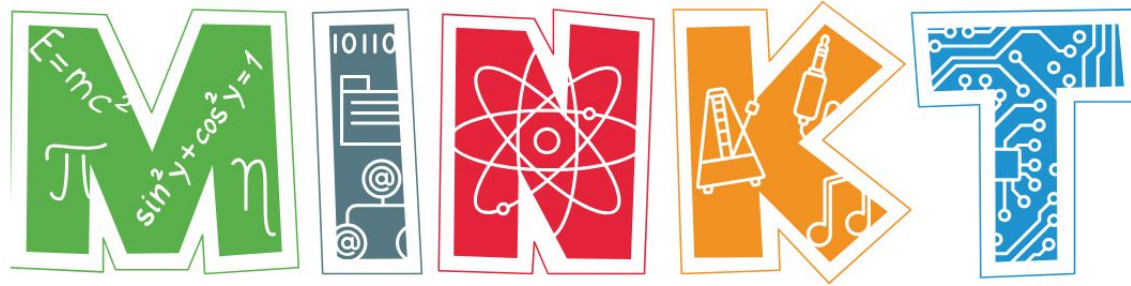


Real-Time sEMG Control of a Bionic Hand for Educational Demonstration

Lukas Zach

Bachelor Thesis Presentation
TU Graz, Institute of Biomedical Imaging
01.12.2025

2 Motivation



Mathematik • Informatik • Naturwissenschaften • Kunst • Technik

LABOR

Motivation



Motivation



Motivation



Motivation



Motivation



Motivation



Goal



Low budget



Representative

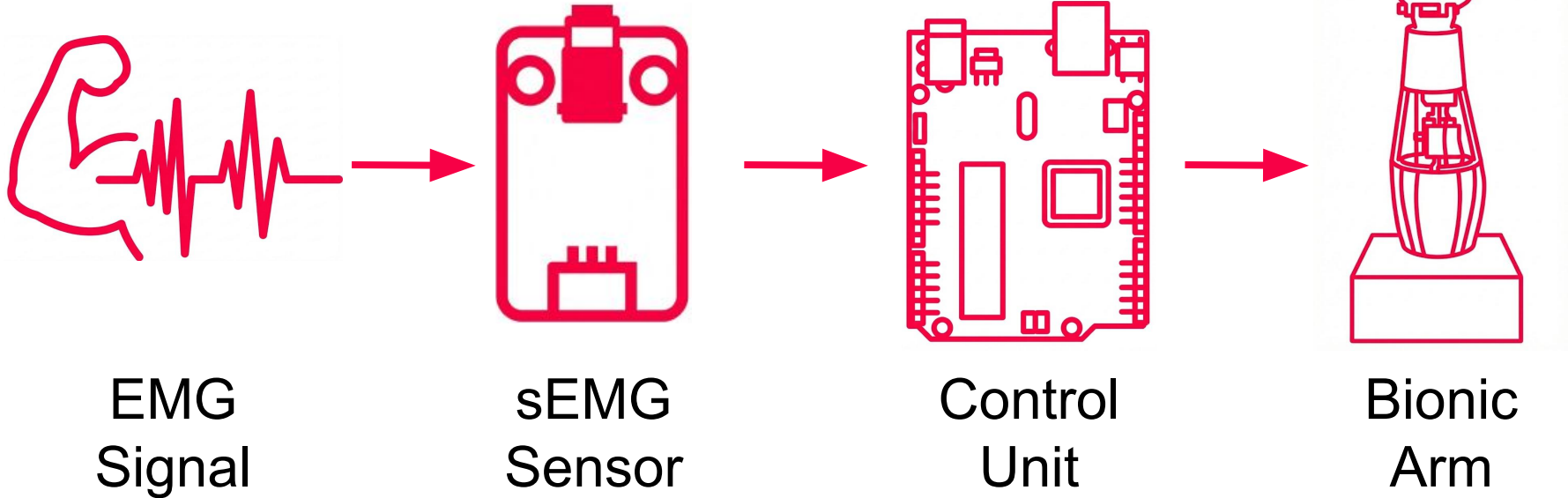


Functioning
Prototype

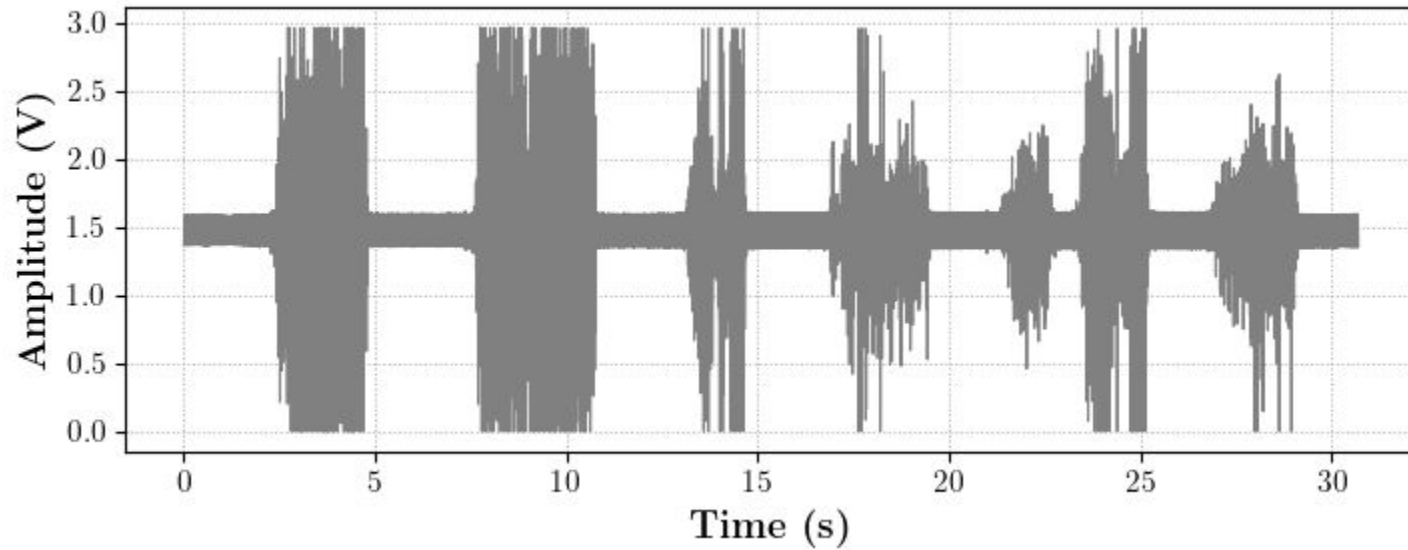


For all age
Groups

System Overview

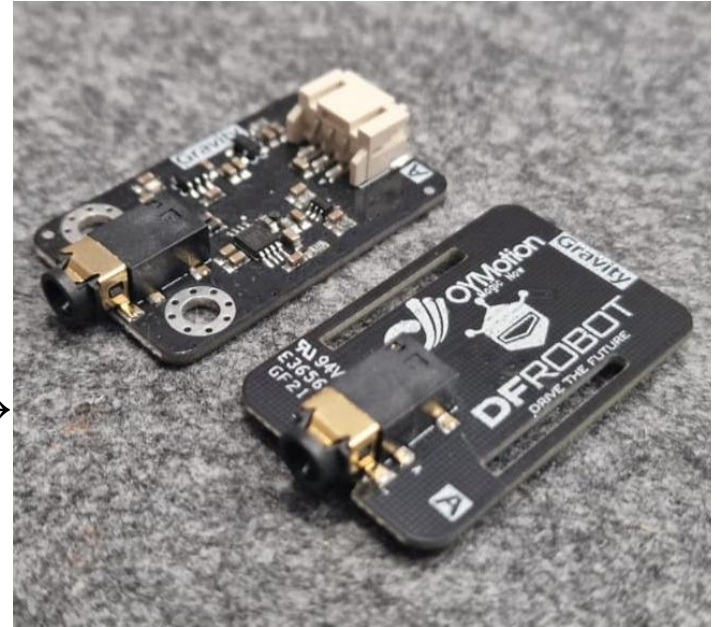


sEMG Signal



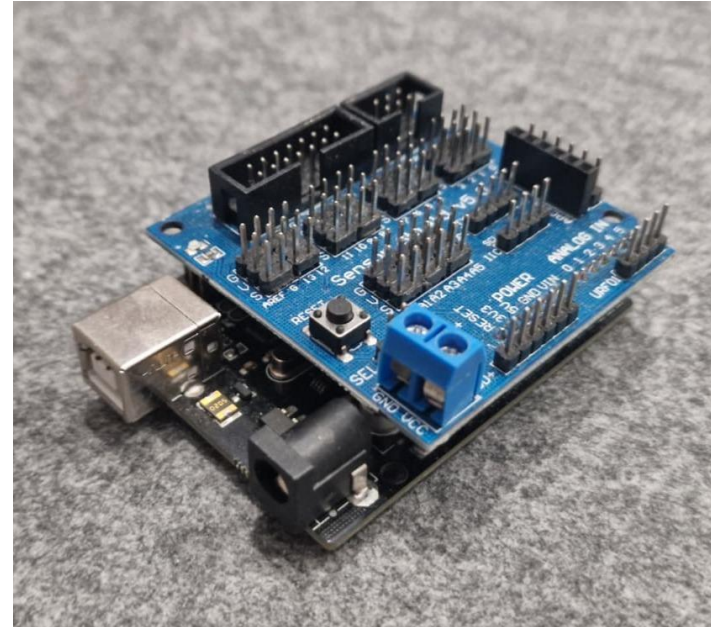
Gravity: Analog sEMG Sensor

- **Type:** Dry Electrode Sensor.
- **Why:** No conductive gel required (ideal for high-traffic lab use).
- **Configuration:** Differential amplification (rejects common noise).
- **Output:** Amplified analog signal (0–3V) → Read by Arduino.

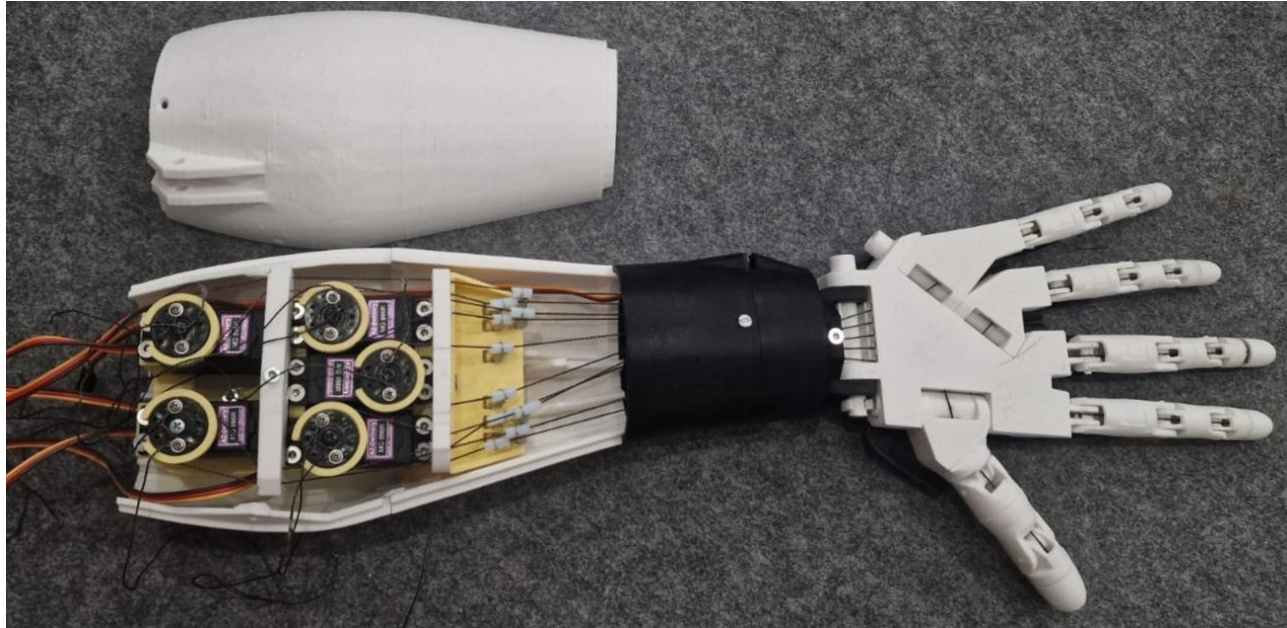


Control Unit

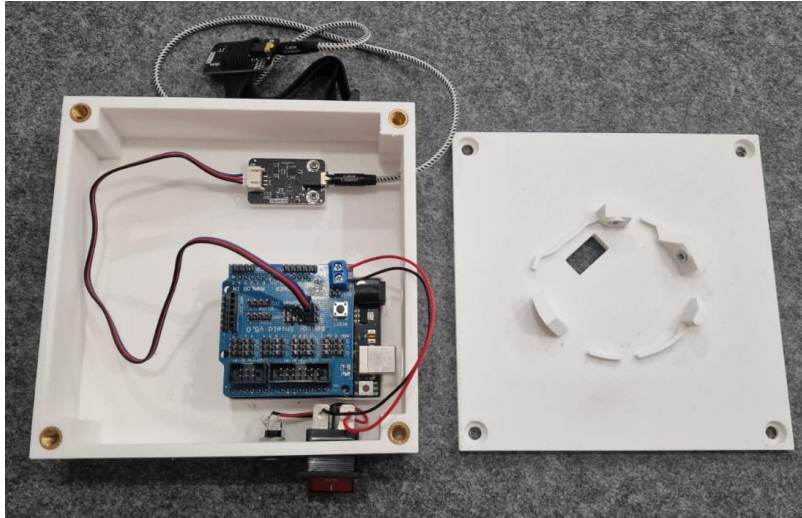
- **Processor:** Arduino Uno Rev3 (Sampling at 1000 Hz).
- **Interface:** Sensor Shield V5.0 for easy component connection.
- **Power Logic:** Isolated Power Supply.
 - Arduino powered via USB.
 - Servos powered by external 15A supply.



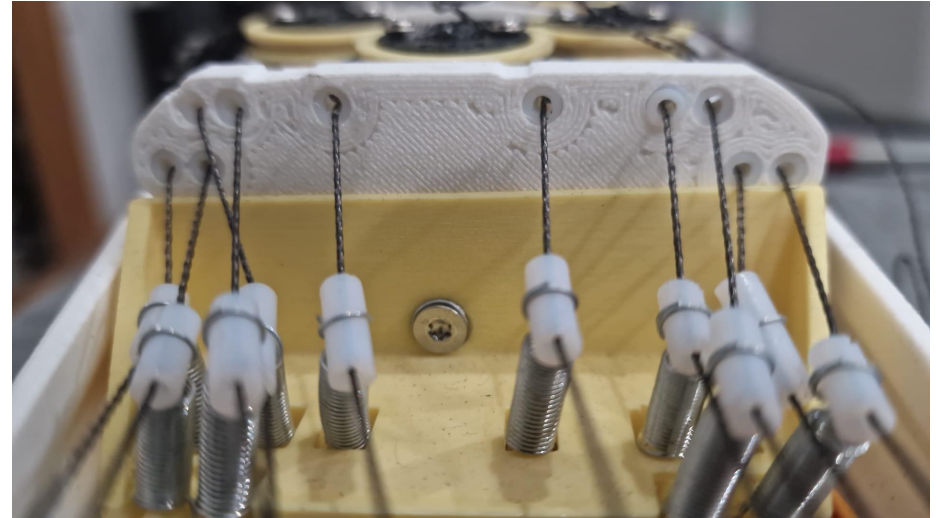
InMoov Bionic Arm



Mechanical Implementation

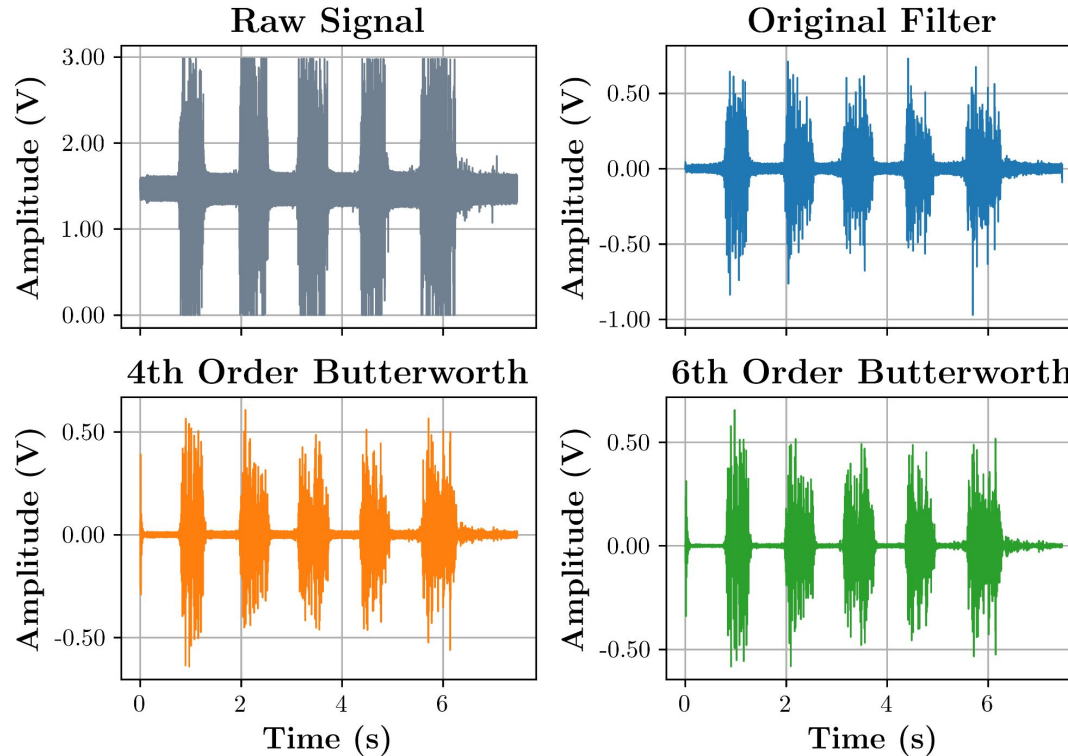


Custom Stand

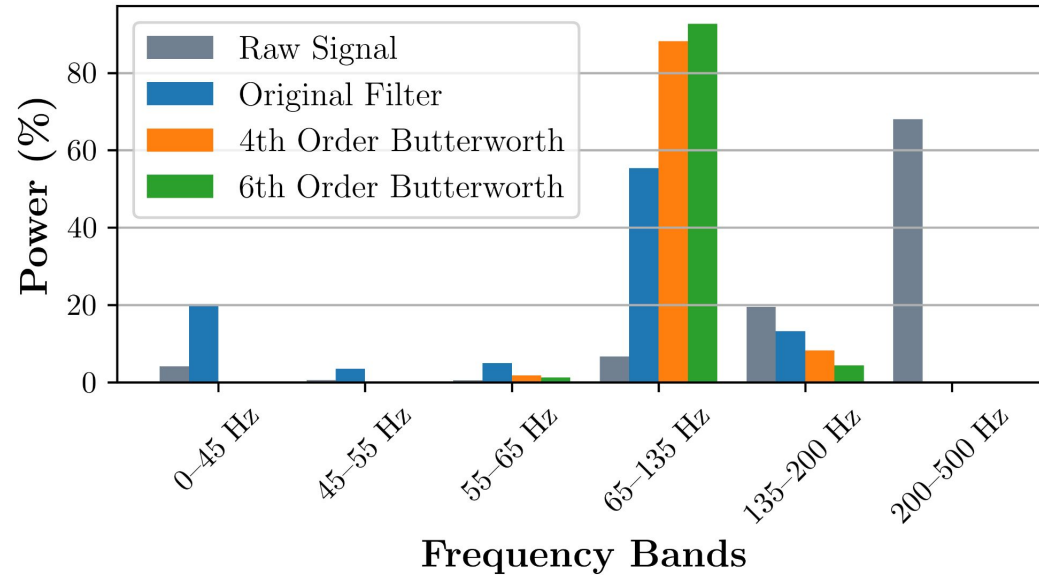


Teflon Enhancements

Signal Processing: Filter Design



Signal Processing: Filter Design

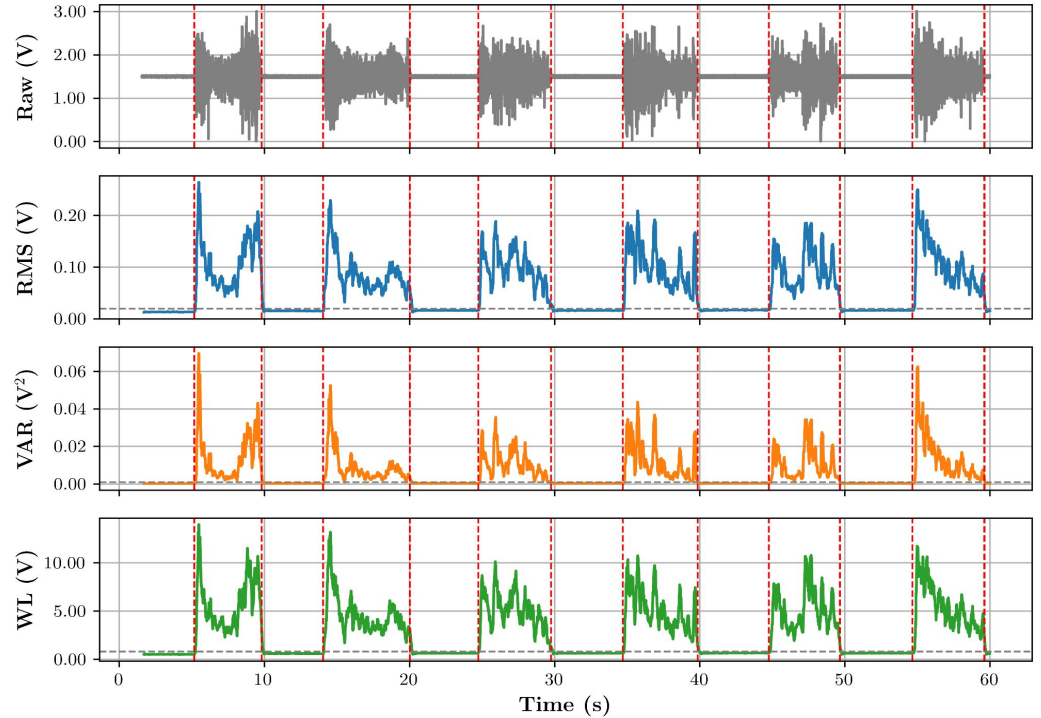


Signal Processing: Feature Extraction

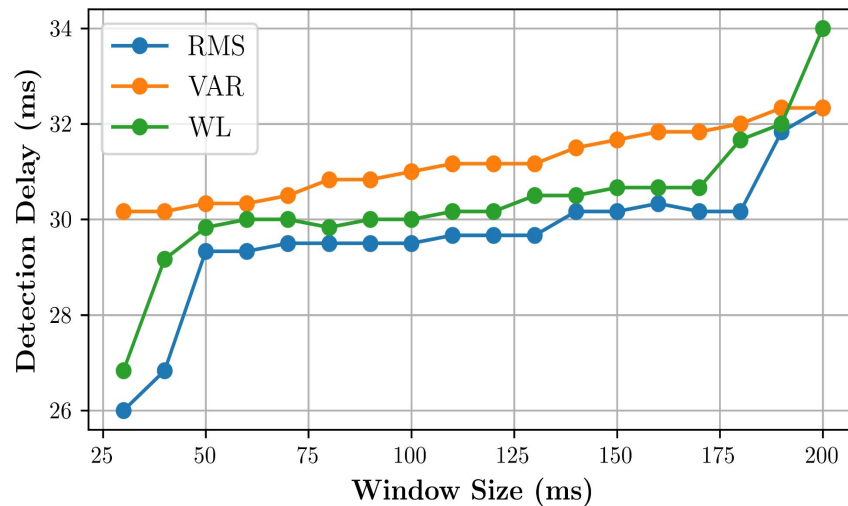
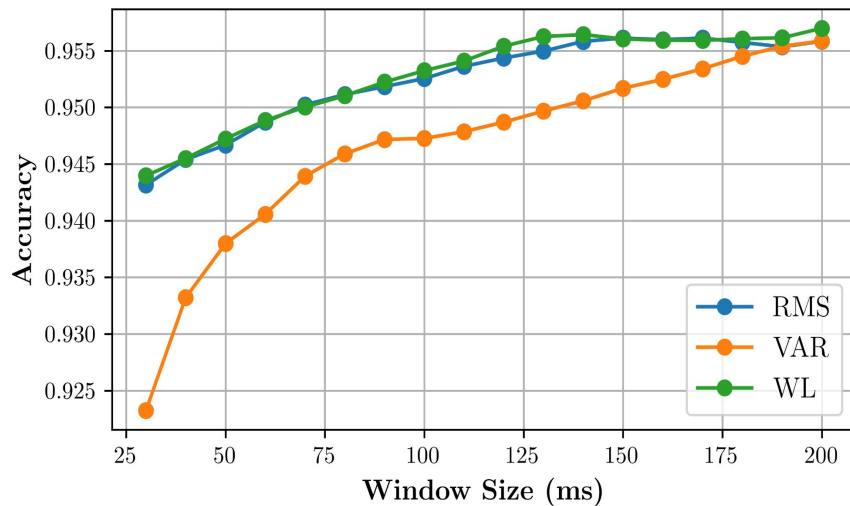
$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

$$\text{VAR} = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

$$\text{WL} = \sum_{i=1}^{N-1} |x(n+1) - x(n)|$$



Signal Processing: Accuracy & Delay



Discussion & Limitations

Signal Stability

- **Issue:** Sensor detachment creates massive noise spikes.
- **Impact:** False positives (unintended closing).
- **Status:** Software "Guard" added, but hardware connection needs improvement.

Mechanical Wear

- **Issue:** Tendons (fishing line) stretch over time.
- **Impact:** Incomplete finger extension.
- **Status:** Teflon tubes reduced friction, but regular re-tensioning is still required.

Usability & Scope

- **Issue:** Manual threshold turning needed per user.
- **Impact:** Validated on N=4 subjects.
- **Status:** Auto-calibration required for unsupervised public use.

Conclusion



Achievements

- ✓ **Functional Prototype:**
Successfully integrated InMoov hand with custom sEMG control.
- ✓ **Safety First:**
Implemented dual-threshold "Disconnection Guard" to prevent erratic movement.
- ✓ **Cost-Effective:**
Created a repairable, modular system suitable for the MINKT Lab.



Futur Work

- **Auto-Calibration:**
Find a algorithm to adjust the thresholds automatically.
- **Long-Term Durability:**
Conduct longer stress tests on the system and improve the tensioning system.
- **Hybrid Features:**
Investigate combining RMS and Waveform Length or other features.

Conclusion



Futur Work

→ Auto-Calibration:

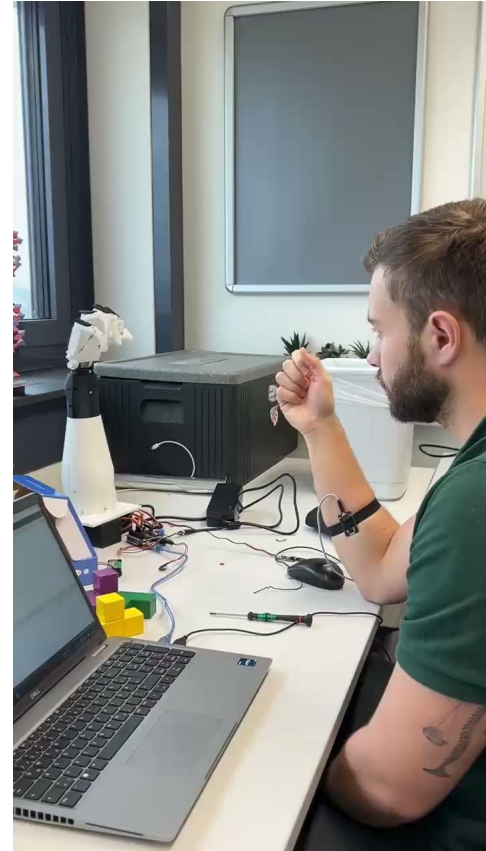
Find a algorithm to adjust the thresholds automatically.

→ Long-Term Durability:

Conduct longer stress tests on the system and improve the tensioning system.

→ Hybrid Features:

Investigate combining RMS and Waveform Length or other features.



Questions?

Thank you for your attention!

Lukas Zach

Bachelorprojekt
TU Graz, Institute of Biomedical Imaging
01.12.2025

References

- **TU Graz Super Science Team.** (n.d.). *TU Graz Super Science Space - MINKT Labor* [Photograph]. Retrieved from <https://super-science-team.tugraz.at/kurse/tu-graz-super-science-space-minkt-labor/>
- **TU Graz Super Science Team.** (n.d.). *Geoday 2023* [Photograph]. Retrieved from <https://super-science-team.tugraz.at/geoday-2023/>
- **Lunghammer.** (n.d.). *Geoday 2023* [Photograph]. Retrieved from <https://super-science-team.tugraz.at/geoday-2023/>
- **Nina Botthof.** (n.d.). *TU Graz Super Science Space - MINKT Labor* [Photograph]. Retrieved from <https://super-science-team.tugraz.at/open-your-mint/>
- **Science Garden.** (n.d.). *Super Science Space - MINKT Labor*. Retrieved [Photograph], from <https://www.sciencegarden.at/mint-erlebnis/super-science-space-minkt-labor-2/>
- *All icons used in this presentation were made using Google Gemini 3.*